Construction and Validation of a Compact Mechatronic Ball-and-Beam Platform for Control Education

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***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Ball-and-beam systems constitute one of the most extensively used benchmark platforms in automatic control education and experimental dynamics research. Their open-loop instability and coupled translational-rotational dynamics make them an ideal testbed for the study of stabilization, trajectory tracking, system identification, and state-feedback regulation [1], [2]. Because of these characteristics, such platforms have been widely adopted in undergraduate laboratories and research environments for the validation of classical controllers, modern optimal strategies, fuzzy logic, sliding-mode, and machine-learning-based approaches [3], [4]. Beyond their pedagogical value, ball-and-beam systems serve as a rigorous case study for dynamic modeling, since their equations of motion, derivable through Newtonian or Lagrangian formulations, yield a double-integrator transfer function whose validation against physical hardware is a non-trivial and meaningful exercise [3].

Despite their conceptual simplicity, commercially available ball-and-beam platforms impose significant acquisition costs that frequently limit access in institutions with constrained budgets. Moreover, commercial solutions typically operate as closed systems, offering limited flexibility for hardware modification, low-level controller programming, or integration of custom sensing and actuation elements [5], [6]. As a result, the literature contains numerous proposals for custom educational prototypes intended to reduce cost and broaden accessibility [5], [7]. However, a recurring limitation across these efforts is the insufficient attention paid to mechanical robustness and long-term experimental repeatability: many prototypes present fragile or ad-hoc mechanical architectures that complicate reproducibility across laboratory sessions or different institutions.

The increasing prevalence of blended and hybrid learning modalities in higher education has highlighted the importance of portable, deployable laboratory tools that students can operate outside dedicated facilities [6], [8]. Portable control platforms enable project-based learning experiences in which students engage with the full engineering cycle—from system design and fabrication to embedded implementation and experimental analysis, without depending on permanently installed laboratory infrastructure [5], [9]. This approach has been shown to improve student engagement and deepen conceptual understanding, particularly in mechatronics and control systems courses [8], [9]. Nevertheless, achieving portability without sacrificing mechanical integrity or experimental fidelity remains an insufficiently addressed design challenge in the existing literature.

A. Related Work

Numerous works have proposed educational ball-and-beam implementations with emphasis on controller development and algorithm evaluation. Recent studies have progressively focused on improving model fidelity and addressing practical limitations commonly neglected in simplified educational implementations. A cascaded LQR-FLC design with hardware-in-the-loop validation was presented in [2], demonstrating that structured cascaded architectures can effectively compensate for model uncertainty arising from incomplete parameter knowledge. A systematic comparative study of ball-and-beam dynamic models including eccentric attachment and friction effects was conducted in [3], where each formulation was experimentally validated in Simulink. The results revealed that standard second-order approximations frequently fail to capture real-system behavior, highlighting the importance of extended and experimentally identified models. Practical actuator limitations were further addressed in [10] through observer-based positioning schemes incorporating motor speed and acceleration dynamics, confirming that neglecting such effects may significantly degrade closed-loop performance in DC-motor-driven implementations.

From the control perspective, Chaturvedi et al. [4] demonstrated through Teaching–Learning-Based Optimization that cascade PID structures tuned with evolutionary algorithms yield superior transient responses and improved robustness against gearbox efficiency variations compared to conventional tuning—an important finding given that low-cost DC motors introduce load-dependent nonlinearities not captured by simplified models. Srivastava et al. [11] similarly applied hybrid optimization to PID design for the ball-and-beam system, reporting consistent improvements in settling time and overshoot rejection. At the platform design level, Takács et al. [5] introduced BOBShield, an open-source miniature ball-on-beam device based on Arduino-compatible hardware and entirely 3D-printed structural components. While the BOBShield demonstrates impressive compactness and cost reduction, its servo-driven actuation and minimal structural mass limit the range of dynamics available for control experimentation. The ball-plate platform described by Knuplez et al. [9] offers a two-dimensional extension using DC servomotors and computer-vision feedback, achieving full mechatronic integration for STEM education; however, the reliance on camera-based sensing and the absence of an experimentally validated dynamic model restrict its applicability to quantitative model-based control studies.

From the control perspective, several studies have researched optimization-based approaches to improve transient performance and robustness. Cascade PID structures tuned using evolutionary algorithms proved to provide improved transient response and enhanced robustness against gearbox efficiency variations compared with conventional tuning approaches in [4] through Teaching-Learning-Based Optimization that. This finding is particularly relevant because low-cost DC motors often introduce load-dependent nonlinearities that are not represented in simplified system models. Likewise, hybrid optimization techniques were applied to PID design for ball-and-beam systems by [11], reporting consistent improvements in settling time and overshoot reduction.

At the educational hardware level, research efforts have increasingly focused on developing compact and accessible experimental platforms. BOBShield, an open-source miniature ball-on-beam device based on Arduino-compatible hardware and fully 3D-printed structural components, was introduced by [5]. While this platform demonstrates considerable advantages in compactness and cost reduction, its servo-driven actuation and reduced structural mass may limit the range of dynamic behaviors available for control experimentation compared with DC-motor-driven implementations. Similarly, the ball-plate platform proposed in [9] extended the concept toward two-dimensional motion through computer-vision feedback and DC servomotor actuation, achieving complete mechatronic integration for STEM education. However, the reliance on camera-based sensing and the lack of experimentally validated dynamic models may reduce its suitability for quantitative model-based control studies.

Further examples within the open-source educational hardware ecosystem, including the FlexLab/LevLab [8] and the PL-TOON platform [6], demonstrate that portable and low-cost mechatronic systems can be designed with sufficient fidelity for meaningful control experimentation. Nevertheless, these platforms primarily address magnetic levitation or cooperative control scenarios and therefore do not reproduce the specific coupled dynamics associated with classical ball-and-beam systems that remain fundamental within control engineering curricula.

Across the broader ball-and-beam literature, a consistent pattern emerges: studies emphasizing advanced control strategies frequently rely on commercial or simplified experimental setups [2], [10], whereas studies prioritizing hardware accessibility often provide limited dynamic model validation and reduced integration between mechanical and embedded subsystems [5], [9]. A synthesis of the reviewed literature further reveals several recurring limitations that frequently appear simultaneously: (i) simplified second-order models assuming instantaneous beam-angle actuation, thereby neglecting the transient lag introduced by DC motors and gearbox transmissions [3], [10]; (ii) mechanical designs emphasizing compactness and cost reduction, which may compromise structural rigidity, repeatability, and long-term robustness [5]; and (iii) limited availability of fully integrated mechatronic architectures combining mechanical design, sensing, custom electronics, and embedded control into compact and reproducible platforms [8], [9]. Consequently, despite substantial progress in controller design and educational hardware development, a gap remains between accessible low-cost platforms and experimentally validated mechatronic systems capable of accurately reproducing practical dynamic behavior while preserving portability, subsystem integration, and reproducibility.

The remainder of this paper is organized as follows. Section II describes the design of the different subsystems, alongside the manufacturing process and chosen components. Section III presents the system variables, assumptions, and mathematical dynamic model of the ball-and-beam. Section IV reports the experimental validation of the two proposed models quantitaively and qualitatively. Section V briefly explains the used cascade control architecture on the actuator. Section VI discusses and analyses the experimental results. Finally, Section VII summarizes the main contributions and outlines future work.

II. System Design and Development

A. Mechanical Design

1) Conceptual Design: The mechanical design of the proposed ball-and-beam platform was driven by three primary objectives: portability, low-cost fabrication, and mechanical repeatability for experimental validation. The system was therefore designed to minimize component count and manufacturing complexity while maintaining sufficient structural rigidity for repeated laboratory operation.

Conventional ball-and-beam systems commonly employ double-lever mechanisms to generate beam inclination. Preliminary evaluations of this configuration revealed increased friction and mechanical play caused by multiple joints and contact interfaces, which can degrade dynamic repeatability and complicate model validation. Consequently, an eccentric-disk mechanism was adopted to directly convert the continuous rotational motion of the actuator into beam oscillation with fewer mechanical interfaces and reduced backlash. In addition, a compression spring was added to the end of the beam to dampen the motion.

Since the selected DC motor operates at relatively high rotational speed, an additional transmission stage was required to increase torque and improve angular resolution at the beam. Although gear-based reductions were initially considered, a belt-and-pulley transmission with a 4:1 reduction ratio was selected due to its smoother operation and reduced backlash. An adjustable rail mechanism was incorporated to maintain belt tension and facilitate shaft alignment.

The final transmission system consists of a motor coupled to a primary shaft supporting the larger pulley. A secondary parallel shaft contains the smaller pulley and an eccentric disk connected through a flange. Both shafts are supported by bearing housings mounted on orthogonal rail structures that allow positional adjustment during assembly.

The beam is fixed at one end and includes a rolling contact element located approximately 28.4 cm from the pivot point, which interfaces with the eccentric disk through a bearing coupling. This arrangement enables smooth motion transmission while minimizing frictional effects.

The complete prototype is integrated within a (35×40) cm structure containing the mechanical assembly, control electronics, and power supply, resulting in a compact and portable experimental platform.

2) 3D Modeling: The complete mechanical assembly was developed using CAD software prior to fabrication. Individual components were designed considering manufacturability, assembly simplicity, and dimensional compatibility between interfacing elements.

Tolerance considerations were incorporated during the design stage to reduce alignment errors and minimize post-processing during assembly. The CAD environment additionally enabled verification of shaft positioning, transmission alignment, and component interference before fabrication.

An exploded view of the final assembly is presented in Fig. 1, while the corresponding system components are summarized in Table I.

3) Manufacturing Process: The prototype combines laser-cut, machined, additive-manufactured, and commercial components in order to balance fabrication simplicity, mechanical performance, and cost.

The beam and beam supports were laser-cut from acrylic to provide a smooth rolling surface and reduce friction between the ball and the beam. Transmission shafts were manufactured from AISI 1020 steel to ensure adequate rigidity, while the motor coupling was machined from brass.

Structural components, including the rails, bearing housings, motor holder, eccentric disk, and rolling coupling, were fabricated in PETG using fused-filament fabrication. The bearing housings were dimensioned according to the specific bearing employed in each stage: housings 1 and 3 support 695 bearings, whereas housings 2 and 4 support MR84 bearings.

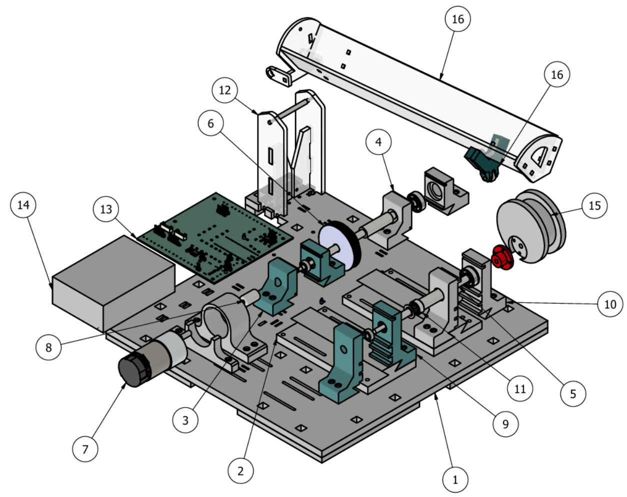

Fig. 1. Exploded view of the proposed ball-and-beam platform.

TABLE I
LIST OF COMPONENTS

Item	Qty	Part Number
1	1	Base
2	2	Rail
3	1	Bearing housing 1
4	1	Bearing housing 2
5	3	Shafts
6	1	Pulley GT2_80T_8mm
7	1	Motor Housing & Holder
8	1	Coupling
9	1	Bearing housing 3
10	1	Bearing housing 4
11	1	Pulley GT2_20T-sh8
12	2	Beam Back Supports
13	1	PCB
14	1	12 V Power Source
15	1	Eccentric Disk and Flange
16	1	Beam
17	1	Rolling Coupling

Commercially available components were used for the pulleys, bearings, flange, and power supply.

B. Electronic System

The electronic system was designed to ensure reliable signal acquisition, robust actuation, and straightforward replication, while maintaining compatibility with portable and low-cost laboratory environments. A custom printed circuit board (PCB) integrates sensing, actuation, and control functionalities into a single platform, reducing wiring complexity and improving system robustness. The PCB supports DC motors with quadrature encoders, digital and analog inputs, as well as PWM and digital outputs, while optocoupling is incorporated to enhance signal integrity and protect low-voltage control electronics from electrical noise and transients generated by the actuator stage, a consideration particularly relevant in educational settings with repeated use and varying operating conditions. Ultimately, the electronic architecture prioritizes modularity and reproducibility, enabling assembly with commonly available components while maintaining sufficient fidelity for experimental validation and control implementation.

1) Sensing: Ball position is measured using an infrared distance sensor (GP2Y0A41SK0F), selected for its compact form factor, low power consumption, and suitability for short-range measurements. The sensor operates within a range of 4-30 cm, which aligns with the physical dimensions of the beam. Its analog voltage output is interfaced directly with the microcontroller through the PCB's analog input channels.

2) Actuation: The actuation subsystem is based on a 12 V DC motor (Pololu) with a nominal speed of 100 RPM and an integrated gearbox with a reduction ratio of 102.08:1. This configuration provides sufficient torque to manipulate the beam while maintaining smooth and controlled motion.

A quadrature encoder with 64 ticks per revolution is integrated into the motor assembly, enabling the estimation of angular position and velocity. These measurements are used both for feedback control and for capturing actuator dynamics. The motor is driven via PWM signals generated by the microcontroller and routed through the PCB.

3) Control Interface: The control interface is implemented using a Raspberry Pi Pico microcontroller, chosen for its balance between computational capability, cost, and widespread availability. The microcontroller handles sensor acquisition, encoder decoding, and actuator command generation, serving as the central integration point of the electronic system.

C. System Integration

The experimental platform integrates the previously described mechanical and electronic subsystems into a compact architecture intended for control experimentation and model validation.

The complete system includes the ball-and-beam mechanism, the actuator and transmission stages, the sensing subsystem, the custom PCB, and the embedded control unit. The platform operates as a self-contained experimental setup capable of signal acquisition, actuator control, and real-time data processing without requiring external laboratory instrumentation.

The modular organization of the components facilitates assembly, maintenance, and future modifications while preserving mechanical stability during operation. Additionally, the compact dimensions of the prototype allow straightforward transportation and deployment in educational and laboratory environments.

The final implementation of the experimental platform is shown in Fig. 2.

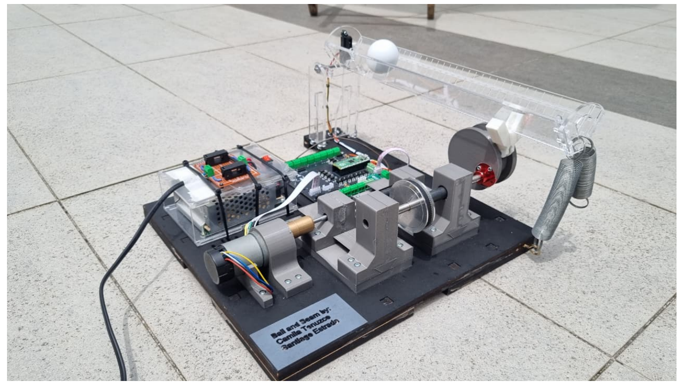

Fig. 2. Experimental ball-and-beam platform used for model validation and control implementation.

III. Dynamic Modeling

A. System Variables

The dynamic model of the ball-and-beam system is derived using the Lagrangian formulation, which is well suited for systems exhibiting coupled translational and rotational motion. This approach enables a systematic derivation of the equations of motion based on energy considerations.

A single generalized coordinate, $r(t)$, is selected to describe the system configuration, representing the linear position of the ball along the beam measured from the pivot point. The inclination of the beam is not treated as an independent generalized coordinate; instead, it is expressed as a function of the motor shaft position. Let $\theta(t)$ denote the angular position of the motor shaft and $\alpha(t)$ the beam inclination angle. These variables are related through a constant gain

$$\alpha(t) = K\,\theta(t), \tag{1}$$

where K is determined experimentally. For the implemented system, $K < 0$, indicating that positive motor rotations produce negative beam inclinations.

A consistent sign convention is adopted in which $r(t) > 0$ corresponds to ball displacement from the pivot toward the motor side, $\theta(t) > 0$ denotes positive motor rotation, and $\alpha(t) > 0$ represents a beam inclination opposite to the direction of increasing $r(t)$. Under this convention, a positive motor angle results in a negative beam inclination, leading to a positive acceleration of the ball toward the motor side.

Within this framework, the mechanical model considers the pulley angle $\theta(t)$, acted on through the motor, as the system input and the ball position $r(t)$ as the output. The model parameters are defined as follows: m is the mass of the ball, J its moment of inertia about the center of mass, δ its radius, and g the gravitational acceleration, treated as a positive scalar constant.

B. Model Assumptions

In order to obtain a tractable analytical model while preserving the dominant system dynamics, several assumptions are introduced. The ball and beam are modeled as rigid bodies, neglecting structural deformations, compliance, and backlash in the transmission mechanism. Furthermore, the inertia of the beam is assumed negligible due to the relatively low beam mass compared with the effective actuator torque capability, such that its motion is entirely determined by the actuator through a fixed kinematic relationship between motor position and beam inclination.

The ball is assumed to roll without slipping along the beam, establishing a kinematic constraint between its translational and rotational motion. Friction is therefore considered sufficient to prevent slipping, while its dissipative effects, along with rolling resistance and aerodynamic drag, are neglected. The kinetic energy expression neglects velocity coupling terms associated with beam angular motion, assuming that beam angular velocities remain sufficiently small relative to the ball translational dynamics.

The governing equation retains the nonlinear gravitational term $\sin\alpha$ prior to linearization. However, for control-oriented analysis, small beam inclinations are considered, allowing the approximation $\sin\alpha \approx \alpha$.

C. Energy Formulation

The dynamic model is obtained using an energy-based approach. The total kinetic and potential energies of the ball are expressed in terms of the generalized coordinate $r(t)$ and its time derivative.

The kinetic energy of the ball consists of both translational and rotational components. The translational kinetic energy is given by $\frac{1}{2}m\dot{r}^2$, while the rotational kinetic energy arises from rolling motion. Under the rolling without slipping assumption, the angular velocity satisfies $\omega = \dot{r}/\delta$, resulting in a rotational kinetic energy of $\frac{1}{2}J\left(\dot{r}/\delta\right)^2$. Combining both contributions, the total kinetic energy can be written as

$$T = \frac{1}{2}\left(m + \frac{J}{\delta^2}\right)\dot{r}^2. \tag{2}$$

The potential energy is determined by the vertical displacement of the ball with respect to the pivot level, which is taken as the reference. From the system geometry, the height of the ball is given by $h = r\sin\alpha$, yielding the gravitational potential energy

$$V = mgr\sin\alpha. \tag{3}$$

The Lagrangian of the system is then defined as

$$L = T - V = \frac{1}{2}\left(m + \frac{J}{\delta^2}\right)\dot{r}^2 - mgr\sin\alpha. \tag{4}$$

D. Equations of Motion

The equations of motion are obtained by applying the Euler–Lagrange equation to the Lagrangian derived in the previous subsection. For the generalized coordinate $r(t)$, this is given by

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0. \tag{5}$$

From the expression of the Lagrangian, the required derivatives are

$$\frac{\partial L}{\partial \dot{r}} = \left(m + \frac{J}{\delta^2}\right)\dot{r}, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = \left(m + \frac{J}{\delta^2}\right)\ddot{r}, \tag{6}$$

and

$$\frac{\partial L}{\partial r} = -mg\sin\alpha. \tag{7}$$

Substituting into the Euler–Lagrange equation yields the nonlinear equation of motion

$$\left(m + \frac{J}{\delta^2}\right)\ddot{r} + mg\sin\alpha = 0. \tag{8}$$

For control-oriented analysis, a linear approximation is obtained by considering small beam inclinations such that $\sin\alpha \approx \alpha$. Using the kinematic relationship $\alpha(t) = K\theta(t)$, the equation becomes

$$\ddot{r}(t) = -\frac{mgK}{m + \frac{J}{\delta^2}}\,\theta(t). \tag{9}$$

Taking the Laplace transform under zero initial conditions, the relationship between the ball position and motor angle is given by

$$s^2R(s) = -\frac{mgK}{m + \frac{J}{\delta^2}}\,\Theta(s), \tag{10}$$

from which the transfer function is obtained as

$$\frac{R(s)}{\Theta(s)} = -\frac{mgK}{m + \frac{J}{\delta^2}} \cdot \frac{1}{s^2}. \tag{11}$$

E. Inclusion of Actuator Dynamics

Classical ball-and-beam models assume instantaneous actuation of the beam angle, resulting in a second-order relationship between the ball position and the input. However, in practical implementations, the actuator exhibits finite response time, introducing additional dynamics that affect system behavior.

The mechanical model first considers the pulley angle θ as the direct system input. Subsequently, the experimentally identified closed-loop actuator dynamics relating the reference angle θ_{ref} to the measured pulley angle θ are incorporated.

To account for this, the closed-loop actuator subsystem is modeled as a first-order system relating the reference angular position to the measured pulley angle. The actuator dynamics are described by

$$G_a(s) = \frac{\Theta(s)}{\Theta_{\text{ref}}(s)} = \frac{k_a}{\tau s + 1}, \tag{12}$$

where $\Theta_{\text{ref}}(s)$ is the reference angle, $\Theta(s)$ is the pulley angle, k_a is the steady-state gain, and τ is the time constant. These parameters are identified experimentally.

By cascading the actuator dynamics with the previously derived ball-and-beam model, the overall transfer function from the reference input to the ball position is obtained as

$$\frac{R(s)}{\Theta_{\text{ref}}(s)} = -\frac{mgK}{m + \frac{J}{\delta^2}} \cdot \frac{k_a}{s^2(\tau s + 1)}. \tag{13}$$

This extended model captures the non-instantaneous response of the actuator, resulting in an augmented third-order input-output model. Under proper calibration, the actuator gain k_a is expected to be close to unity, ensuring that the steady-state motor position closely tracks the reference input.

IV. Experimental Validation

A. Excitation Signal

A step excitation signal was used to experimentally evaluate the proposed models. In all experiments, the excitation was applied to the reference pulley angle $\theta_{\mathrm{ref}}(t)$.

The step excitation was defined as

$$\theta_{\mathrm{ref}}(t) = \begin{cases} 0, & t < 0, \\ 30^\circ, & t \geq 0, \end{cases} \tag{14}$$

The step response was used to evaluate the transient behavior of the system under open-loop ball-position conditions while maintaining closed-loop actuator regulation.

B. Experimental Setup

Experiments were conducted using MATLAB and Simulink R2025b. The control and data acquisition routines were implemented in Simulink and deployed to a Raspberry Pi Pico through the Arduino Hardware support package. This approach reduced encoder-sampling inconsistencies.

The cascade control structure operated entirely on the microcontroller. The measured angular position was converted to the pulley angle according to the transmission ratio, such that $\theta_{\mathrm{pulley}} = \theta$. The ball position was measured using the previously described infrared sensor.

The experimental data recorded during the validation process included the reference angle θ_{ref}, the pulley angle θ, and the ball position r. All signals were sampled with a sampling period of

$$T_s = 0.01 \text{ s}, \tag{15}$$

and logged in Simulink using the `To Workspace` block for subsequent processing and analysis in MATLAB.

C. Actuator Identification

In order to account for the non-instantaneous response of the actuator, an experimental identification procedure was carried out for the angular positioning stage, directly from the controlled pulley-angle closed-loop response. Step variations were applied to the reference angle θ_{ref}, and the corresponding pulley angle response θ was recorded using the encoder measurements acquired from the motor shaft.

Multiple step responses were analyzed during the identification process. Among them, the transition from 10° to 30° was selected as a representative operating condition, since it is consistent with the expected angular excursions during ball-and-beam operation. The experimental response was normalized with respect to its initial condition, and a first-order approximation was obtained analytically from the steady-state gain and time constant of the measured response.

The steady-state gain was estimated from the ratio between the output and input variations, while the time constant was determined using the 63% response criterion. To reduce discretization effects associated with the sampling period, the time constant estimation considered neighboring samples around the identified crossing point.

The resulting closed-loop actuator subsystem is given by

$$G_a(s) = \frac{\Theta(s)}{\Theta_{\mathrm{ref}}(s)} = \frac{0.996}{0.425s + 1}, \tag{16}$$

indicating a first-order behavior with a steady-state gain close to unity.

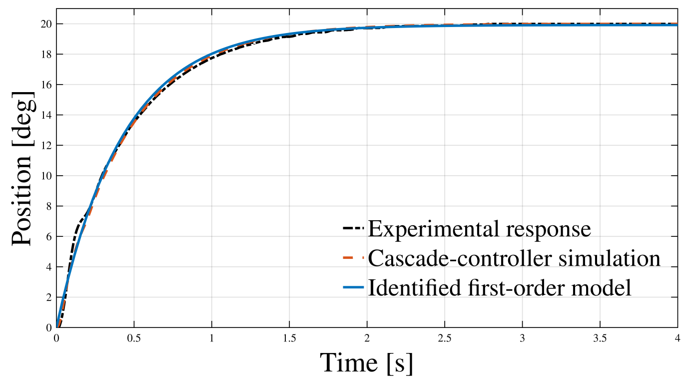

Fig. 3. Comparison between the experimental actuator response, the identified first-order model, and the Simulink simulation of the implemented cascade controller for a representative angular step input.

Fig. 3 compares the experimental response, the identified first-order model, and the Simulink simulation of the implemented cascade controller. The quality of the approximation was evaluated using the normalized percentage fit metric (17), where r is the experimental response, $\hat{r}$ is the estimated response, and $\bar{r}$ is the mean value of the experimental data, yielding values of 95.69% for the cascade-controller simulation and 94.50% for the identified first-order model.

$$\mathrm{Fit}(\%) = 100 \left(1 - \frac{\|r - \hat{r}\|}{\|r - \bar{r}\|}\right) \tag{17}$$

D. Model Validation and Performance Evaluation

The proposed third-order model was experimentally validated through comparison with the real ball-and-beam response under a step excitation described in (14). For simulation purposes, the reference signal was converted to radians in order to maintain consistency with the transfer-function representation. The experimental response was compared against both the classical second-order model, which assumes instantaneous beam-angle actuation, and the proposed third-order model incorporating the identified actuator dynamics (16).

The parameters employed in the simulations are summarized in Table II.

TABLE II
Parameters used for model validation.

Parameter	Value
m	287.77×10^{-3} kg
δ	41.2×10^{-3} m
J	1.9539×10^{-4} kg m^2
K	-0.042
g	9.81 m/s^2

Fig. 4 presents the comparison between the experimental response, the classical second-order model presented in (11), and the proposed third-order model (13). The results show that the second-order model fails to accurately capture the transient lag observed in the physical system due to the assumption of instantaneous beam-angle actuation. In contrast, the proposed third-order model reproduces the experimental behavior more closely by incorporating the experimentally identified actuator dynamics.

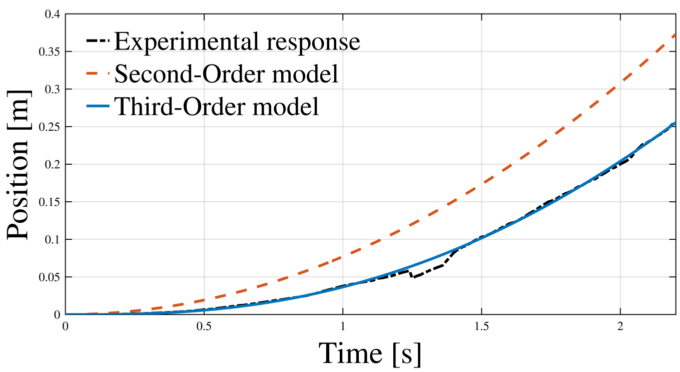

Fig. 4. Comparison between the experimental response, the classical second-order model, and the proposed third-order model under a step excitation.

To quantitatively evaluate model accuracy, the root-mean-square error (RMSE) was computed as

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(r_i - \hat{r}_i)^2}, \tag{18}$$

where r_i represents the experimental ball position and $\hat{r}_i$ corresponds to the estimated response; and the percentage fit followed (17).

The second-order model achieved a normalized percentage fit of 16.64% with an RMSE of 0.06256 m, whereas the proposed third-order model obtained a fit of 94.69% and an RMSE of 0.00398 m. These results demonstrate that incorporating actuator dynamics significantly improves agreement with the observed experimental response within the evaluated operating conditions.

V. Cascaded Control Structure

A cascaded control structure is implemented to ensure actuator-level stability and enable reproducible experimental conditions during model validation. Unlike conventional ball and beam systems that employ servo actuators, the proposed platform uses a low-cost DC motor with pulley reduction, requiring closed-loop regulation of the actuator dynamics prior to ball position control. The controller is therefore introduced exclusively to guarantee consistent angular positioning behavior and reduce the influence of motor nonlinearities and load-dependent disturbances on the experimental results. This additionally promotes repeatable experimental behavior across low-cost educational implementations of the platform.

The cascaded architecture shown in Fig. 5 consists of an outer angular position loop and an inner velocity loop. The outer loop regulates the output shaft angular position θ in

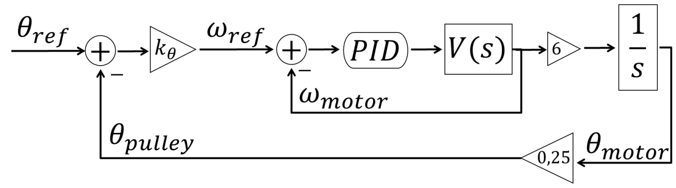

Fig. 5. Implemented cascaded control structure. The outer proportional controller regulates the pulley angle θ_{pulley}, denoted as θ throughout this paper, and generates the velocity reference ω_{ref} for the inner discrete PID controller. Conversion factors account for pulley reduction and unit transformations.

degrees using a proportional controller, which generates the reference signal for the inner loop. The inner loop controls the motor shaft velocity in RPM through a discrete PID controller using encoder-based velocity estimation obtained through discrete differentiation of encoder measurements. This structure is intended to improve disturbance rejection [12], [13] to mitigate velocity variations caused by changes in effective motor inertia [14], allowing the actuator dynamics to exhibit repeatable first-order behavior during experimentation. The control algorithm is implemented digitally on the Raspberry Pi Pico platform, with the outer position loop executed at a sampling period of 0.01 s.

VI. Discussion

A. Comparison with Existing Systems

The proposed platform addresses several of the recurring limitations identified in the reviewed literature regarding educational ball-and-beam systems, particularly those related to mechanical repeatability, subsystem integration, portability, and dynamic model fidelity. Unlike many low-cost educational implementations that prioritize compactness at the expense of structural rigidity [5], the presented prototype was intentionally designed to preserve mechanical robustness while remaining portable and reproducible. The mechanical architecture combines laser-cut acrylic components, machined metallic transmission elements, PETG structural supports, and bearing-guided shafts in order to reduce backlash, improve alignment stability, and maintain repeatable operation across successive experimental sessions. In addition, the adoption of an eccentric-disk actuation mechanism reduces the number of mechanical interfaces compared with conventional lever-based configurations, thereby minimizing frictional effects and mechanical play that may otherwise compromise experimental repeatability and model validation.

From the perspective of portability and educational deployment, the complete system integrates the mechanical assembly, sensing subsystem, custom PCB, embedded controller, and power supply within a self-contained (35×40) cm structure weighing approximately 3 kg. This compact configuration addresses the increasing demand for deployable laboratory platforms suitable for hybrid and project-based learning environments discussed in [6], [8]. In contrast to larger laboratory-oriented commercial systems, the proposed implementation can be transported and operated without requiring external instrumentation or dedicated infrastructure while still preserving sufficient mechanical stability for quantitative experimentation.

The proposed platform additionally emphasizes replicability through the use of commercially available components, additive manufacturing processes, and modular subsystem integration. Unlike commercial educational platforms that frequently operate as closed systems with limited hardware accessibility, the presented architecture enables straightforward reproduction and future modification using accessible fabrication methods and low-cost embedded hardware. The CAD files, PCB design, and implementation resources are available from the authors upon request, facilitating reproduction in educational and research environments.

A further distinction with respect to several existing educational implementations lies in the degree of mechatronic integration achieved by the platform. Previous low-cost systems often focus primarily on either controller development or mechanical accessibility while providing limited integration between sensing, embedded control, actuation, and custom electronics [5], [9]. In contrast, the proposed system integrates the ball-and-beam mechanism, encoder-based actuator feedback, infrared position sensing, embedded cascade control, and onboard power regulation into a unified architecture specifically intended for experimental validation and control implementation.

From the modeling perspective, the experimental results directly support the research gap identified in the literature regarding simplified second-order formulations that neglect actuator dynamics [3], [10]. While classical ball-and-beam models assume instantaneous beam-angle actuation, the implemented DC-motor-driven platform exhibits measurable actuator dynamics that significantly influence the transient response. By experimentally identifying the closed-loop actuator behavior and incorporating it into the system formulation, the proposed third-order model achieved a normalized fit of 94.69% and an RMSE of 0.00398 m, compared with 16.64% and 0.06256 m obtained using the classical second-order approximation. These results demonstrate that incorporating experimentally identified actuator dynamics substantially improves agreement between analytical predictions and physical system behavior within the evaluated operating conditions.

B. Model Limitations

Although the proposed third-order model achieved strong agreement with the experimental response, several sources of deviation remain due to the simplifying assumptions adopted during the modeling process. The formulation neglects dissipative phenomena such as rolling resistance, bearing friction, belt elasticity, and aerodynamic effects, all of which may contribute to small discrepancies between the simulated and measured responses. Likewise, the beam and transmission components were assumed rigid, while practical implementations may exhibit minor compliance, alignment tolerances, and residual mechanical play despite the measures taken to reduce backlash. The infrared sensor additionally introduces measurement noise and nonlinearities associated with distance-dependent sensitivity and surface reflectivity. Furthermore, the actuator dynamics were approximated as a first-order subsystem identified around a representative operating condition, such that variations in load, supply voltage, or motor nonlinearities outside the evaluated range may slightly modify the transient response.

Despite these limitations, the experimental results demonstrate that the proposed formulation captures the dominant dynamics of the physical system with sufficient fidelity for control-oriented analysis and educational experimentation. In particular, the inclusion of experimentally identified actuator dynamics significantly improved the correspondence between the analytical and measured responses compared with the classical second-order approximation.

VII. CONCLUSIONS

This work presents the design, modeling, and experimental validation of a compact and portable ball-and-beam platform intended for control education and research applications. The principal contributions are summarized as follows:

- **Portable and reproducible mechatronic platform:** A self-contained (35×40) cm architecture integrating the ball-and-beam mechanism, belt-and-pulley transmission, custom PCB, and embedded controller into a single transportable unit.
- **Mechanical robustness through deliberate design:** An eccentric-disk transmission mechanism designed to reduce mechanical play and friction compared to conventional lever-arm configurations, combined with laser-cut, machined, and additive-manufactured components selected for structural repeatability.
- **Extended Lagrangian-based dynamic model:** A third-order transfer function derived from the Lagrangian formulation and augmented with experimentally identified first-order actuator dynamics, overcoming the limitations of classical second-order approximations.
- **Rigorous experimental validation:** Quantitative comparison of second- and third-order model predictions against real-system step responses, demonstrating a normalized fit improvement from 16.64% to 94.69% upon inclusion of actuator dynamics.
- **Cascaded control for repeatable operation:** Implementation of an outer proportional position loop and inner discrete PID velocity loop on a Raspberry Pi Pico, ensuring consistent actuator behavior suitable for experimental model validation in educational settings.

The results demonstrate that low-cost and portable experimental platforms can still provide sufficient mechanical repeatability and dynamic fidelity for meaningful control-oriented analysis. By combining experimentally validated modeling, embedded implementation, and integrated mechanical-electronic design within a compact architecture, the proposed system contributes toward reducing the gap between theoretical instruction and practical experimentation commonly encountered in control engineering education. In particular, the platform enables students and researchers to directly observe the influence of actuator dynamics, subsystem integration, and non-ideal physical behavior on real-system responses, aspects that are frequently simplified or omitted in purely simulation-based environments.

The self-contained and transportable architecture further supports deployment in project-based, hybrid, and resource-constrained laboratory environments where permanently installed experimental infrastructure may not be available. Consequently, the proposed system demonstrates that portable mechatronic platforms can simultaneously preserve accessibility, reproducibility, and experimental realism while remaining sufficiently flexible for both instructional and research-oriented applications.

A. Future Work

The presented platform additionally provides a reproducible experimental framework suitable for future research involving system identification, advanced control strategies, state estimation, nonlinear compensation, and embedded real-time implementation. Future work may therefore extend the formulation through the inclusion of friction compensation, nonlinear actuator behavior, beam flexibility effects, or higher-order identification techniques in order to further improve predictive accuracy under broader operating conditions.

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